Homework 4: Human–Machine Attention Alignment Case Study

115-1 Human Perceptual-Centric Vision and Multimodal Representation Learning

Assignment Schedule

Released: November 10, 2026 (Week 10); **Related Lecture:** November 17, 2026 (Week 11) — Explainability and Visual Attention in Human and Machine Vision; **Due:** December 1, 2026 (Week 13), before class.

Students will have approximately **three weeks** to complete the assignment.

1 Objective and Learning Goals

The goal of this assignment is to investigate whether a machine-learning model "looks" at the same visual information as humans when making a prediction.

Students will compare human visual attention, represented by eye-tracking fixations or saliency maps, with model explanations such as Grad-CAM or Attention Rollout. The objective is not only to visualize attention maps, but to **quantitatively evaluate human–model alignment** and identify cases where model explanations differ substantially from human attention.

Human Attention ↔ Model Explanation → Quantitative Alignment → Misalignment Analysis

Students should learn that a visually plausible heatmap does not necessarily imply that a model relies on human-like visual evidence.

2 Task

This homework should be completed in teams of **2–3 students**.

Each team will use a public dataset containing human fixation or saliency annotations and compare them with explanations generated from a pretrained vision model.

Possible human-attention datasets include **SALICON**, **MIT1003**, **CAT2000**, or another appropriate public dataset.

For the machine side, students may use a pretrained CNN, Vision Transformer, CLIP-based image encoder, or another suitable vision model.

Generate model explanations using at least **one** method, such as:

- Grad-CAM or a related gradient-based attribution method;
- Attention Rollout for a Vision Transformer; or
- another spatial explanation method with clear justification.

Training a new model and collecting new eye-tracking data are **not required**.

3 Minimum Requirements

Evaluate human–model attention alignment on at least **50 images** from the selected dataset.

Human and model attention maps should be transformed into a compatible spatial representation before comparison. Students should clearly describe any resizing, normalization, smoothing, or thresholding used in this process.

Use at least **one quantitative alignment metric**. Possible choices include:

- Pearson correlation coefficient (CC);
- Normalized Scanpath Saliency (NSS);
- histogram or saliency-map similarity (SIM);
- KL divergence;
- AUC-based saliency measures; or
- another justified spatial alignment measure.

Using two complementary metrics is encouraged but not required.

In addition to reporting the average alignment score, students must inspect individual cases to determine when human and machine attention agree or disagree.

4 Misalignment Case Study

Identify at least **5 representative cases** with low human–model attention alignment.

For each case, present the input image, human attention map, model explanation, and alignment score. Analyze what visual evidence the human observer and model appear to prioritize differently.

Possible explanations include background context, object texture, edges, small salient objects, faces, text, contextual shortcuts, or other task-irrelevant regions.

When relevant, students should consider whether the observed behavior suggests a possible **shortcut** or a failure of perceptual alignment.

The main question is:

Is the model using visual evidence that is consistent with human attention?

5 Required Content

The report should contain the following components.

1. Dataset and Model Briefly describe the human-attention dataset, the selected image subset, the pretrained model, and the explanation method.

2. Experimental Setup Explain how human attention maps and model explanation maps are generated and processed before comparison. Describe the selected alignment metric and why it is appropriate for the analysis.

3. Quantitative Alignment Report the distribution or average human–model alignment score using at least one quantitative metric.

Include at least one figure or table summarizing the overall results.

4. Misalignment Cases Present at least **5 representative low-alignment cases**, showing the input image, human attention, and model explanation together.

Explain the likely reason for each disagreement rather than simply stating that the heatmaps are different.

5. Discussion Discuss what the results suggest about the relationship between human visual attention and model explanations.

Consider whether attention alignment changes with image content, model correctness, salient objects, or other observable factors. Identify at least one limitation of the current analysis and one possible follow-up experiment.

6 Practical Notes

Do not train a large model from scratch. Using pretrained models and existing explanation implementations is strongly recommended. The focus of this assignment is analysis rather than model training.

Do not evaluate alignment only by visual inspection. Heatmaps can appear similar even when their spatial distributions differ substantially. At least one quantitative alignment measure is required.

Attention or attribution is not automatically an explanation of model reasoning. Students should interpret Grad-CAM, attention maps, and related methods carefully and avoid claiming that they directly reveal the complete internal decision process of the model.

7 Submission

Submit:

(1) Report: A concise PDF describing the dataset, model, alignment analysis, misalignment cases, and interpretation.

(2) Code: Scripts or notebooks required to generate model explanations, compute alignment scores, and reproduce the reported figures.

The recommended report length is **3 pages**, with an absolute maximum of **4 pages**, excluding references and the Contribution Statement.

Include a short **Contribution Statement** describing each team member's role.

All submissions must follow the course-wide *Academic Integrity and Generative AI Policy*.

8 Grading

Component	Points
Understanding of human attention and model explanation	15
Experimental setup and reproducibility	15
Quantitative attention-alignment analysis	20
Misalignment case selection and visualization	20
Perceptual interpretation and critical analysis	20
Clarity, organization, and presentation	10
Total	**100**

A strong submission should go beyond producing visually appealing heatmaps. The analysis should use quantitative evidence to identify **where, when, and why** human and machine attention differ.